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Formulation and evaluation of polyherbal toothpaste

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Abstract
This study focused on the preparation and evaluation of a polyherbal toothpaste formulated from natural ingredients for oral healthcare. The increasing use of synthetic chemicals in commercial toothpastes has raised concerns regarding adverse effects such as tooth sensitivity, irritation, and damage to oral tissues. To overcome these limitations, an herbal toothpaste was prepared using medicinal plants known for their therapeutic value. The formulation contained neem leaves, guava leaves, clove, cardamom, and turmeric along with honey, baking soda, and coconut oil. These ingredients were selected because of their antimicrobial, anti-inflammatory, antioxidant, and antiseptic activities beneficial for maintaining oral hygiene. The plant materials were collected from different areas of Sikkim, shade dried, powdered, and blended in appropriate proportions to prepare the toothpaste. The prepared formulation was subjected to various evaluation tests including pH determination, organoleptic assessment, irritancy test, foaming ability, and viscosity measurement. The toothpaste showed a pH of 7.5, indicating suitability for oral use and compatibility with tooth enamel. Organoleptic evaluation revealed an olive-green appearance with a pleasant odor and slightly bitter taste. The irritancy test showed no signs of redness or irritation, confirming the safety of the formulation. Adequate foam formation demonstrated satisfactory cleansing ability. The viscosity values ranged between 5.07×10^3 and 5.10×10^3 cP, indicating acceptable consistency and smooth extrusion. The findings suggest that the formulated polyherbal toothpaste is safe, stable, economical, and effective for natural oral care.

Keywords: Polyherbal toothpaste, herbal formulation, oral hygiene, antimicrobial activity, physicochemical evaluation, dental plaque

1. Introduction
Oral hygiene plays a vital role in maintaining overall health. Toothpaste is widely used for cleaning teeth, preventing dental caries, and maintaining fresh breath. Conventional toothpaste contains synthetic agents which may cause side effects like irritation, staining, and sensitivity [1]. Herbal toothpaste, prepared using plant-based ingredients, is considered safer and eco-friendly. Traditional medicinal plants such as neem, clove, and tulsi have been used for centuries due to their antibacterial and therapeutic properties [2].
For generations, people have believed that the neem tree (*Azadirachta indica*) possesses amazing health-promoting qualities. In actuality, there is evidence that neem was being used to promote healing as early as 4,500 years ago [3]. In the Indian medical system, neem (*Azadirachta indica*) is a key component of homeopathic and ayurveda remedies as well as traditional healers. Neem is a fearsome natural gift and an omnipotent tree.Before written records were available to document the beginning of history, people employed the herbal neem plant widely to treat a variety of illnesses [4].
Maintaining good oral hygiene is crucial for general health and wellbeing.
As a portal to the human body, the mouth cavity is continuously in contact with food particles, microbes, and ambient elements. Tooth issues such as tooth cavities, gingivitis, periodontitis, foul breath, and plaque buildup can result from poor oral hygiene Dentalcaries and periodontal disorders are the most prevalent chronic ailments that afflict peop le globally [5]. Herbal ingredients such as neem (*Azadirachta indica*), clove (*Syzygium aromaticum*), turmeric (curcuma longa), and guava leaf are well known for their therapeutic properties [6]. Neem possesses strong antibacterial and antifungal activity, making it effective against plaque-forming bacteria. Clove contains eugenol, which provides analgesic and antiseptic effects, helping to relieve toothache. Turmeric exhibits anti-inflammatory and antimicrobial properties, while guava leaf acts as a natural, gentle, and effective supplement to regular oral hygiene [7].

1.1 Herbal toothpaste advantages over synthetic toothpaste

1. **Fewer Side Effect:** Herbal toothpaste is made from natural plant ingredients, so it is less likely to cause irritation, allergies, or tooth plaque [8].
2. **sensitivity compared to synth:** Herbs like neem, clove, and guava leaf have natural antibacterial properties that help control plaque and oral infections without harsh chemicals [9].
3. **No Harmful Chemicals:** It does not contain synthetic substances like triclosan, parabens, or artificial flavors, which may have long-term harmful effects [10].
4. **Eco-Friendly and Biodegradable:** Herbal products are environmentally safe and biodegradable, reducing pollution compared to chemical-based toothpaste [11].

The aim of the research was to formulate an effective herbal toothpaste using natural ingredients and to evaluate physical and chemical properties of it.

2. Materials and Methods

2.1 Plant collection

Collection of Neem (*Azadirachta indica*)

The leaves of neem were collected from Majitar, East Sikkim, after that the leaves were shade dried and coarsely powdered using mortar and pestle.



Fig 1: Neem leave

Collection of Guava leaf (*Psidium guajava*)

The guava leaves were collected from Majitar, East Sikkim, after that the leaves were shade dried and coarsely powdered using mortar and pestle.



Fig 2: Guava leave

Collection of Clove (*Syzygium aromaticum*)

The clove were collected from Kumrek, Rangpo, after that the clove were shade dried and coarsely powdered using mortar and pestle.



Fig 3: Clove (source-India Flora Online)

Collection of Cardamom (*Elettaria cardamomum*)

The cardamom were collected from Singtam, Sikkim, after that the seeds were removed, shade dried and coarsely powdered using mortar and pestle.



Fig 4: Cardamom

Collection of Turmeric (*Curcuma longa*)

The turmeric were collected from local farmer of kumrek, Rangpo, after that it were shade dried and coarsely powered using grinder.



Fig 5: Turmeric (source- India Flora Online)

Collection of Honey, Baking soda, Coconut oil

The honey, baking soda, coconut oil were collected from the local market of Majhitar.



Fig 6: Honey



Fig 7: baking soda



Fig 8: Coconut oil

2.2 Plant materials

The plants are *Azadirachta indica A.Juss* (neem), *Psiduim guajava L* (guava leaves), *Syzyguim aromaticum* (clove buds), *Elettaria cardamomum* (cardamom), *Curcuma longa L* (turmeric powder) were collected, dried and mixed together for preparation of herbal toothpaste.

2.3. Method

1. **Powder Plant Material preparation:** Powder plant was prepared by washing the plants leaves and dried them properly at room temperature and was ground into powder by the mechanical grinder.
2. **Mixing of ingredients:** mixture of powdered herbs such as neem leaves, guava leaves, clove buds, Cardamom seeds, turmeric, honey, baking soda and coconut oil were taken and mixed together as per the following table. (Table no. 1).

Table 1: Formulation table

Serial no	Ingredients	Quantity	Use
1	Neem powder	7.5 gm	Anti-bacterial agent
2	Guava powder	2.5 gm	Anti-microbial agent
3	Clove	2.5 gm	Anti-inflammatory agent
4	Cardamom	2.5 gm	Anti-bacterial and anti-inflammatory agent
5	Turmeric	1 teaspoon	Anti-microbial and anti-inflammatory properties
6	Honey	Quality sufficient	Anti-bacterial and anti-inflammatory properties
7	Baking soda	2 gm	Alkaline properties
8	Coconut oil	2 ml	Anti-microbial agent

Take fresh neem leaves, guava leaves, cloves, and cardamom and dry them at room temperature; then make a fine powder with the help of a mechanical grinder. Weigh 7.5 gm of neem powder and 2.5 gm of guava powder and mix them well, then add 2.5 gm of clove powder and 2.5 gm of cardamom powder. Mix them well to make a paste. Add 2 ml of coconut oil and 1 tablespoon of turmeric powder. Mix them well to make a paste. Then add 2 gm of baking soda and a few drops of honey; mix all the above mixture because honey removes the bitter taste of neem powder. Hence, prepare toothpaste and store it in a well-sealed container [18].



Fig 9: Formulated toothpaste

2.4 Evaluation of formulated herbal toothpaste

2.4.1 pH paper test

The pH test using pH paper is a simple and rapid method to determine the acidity or alkalinity of the toothpaste formulation. Maintaining an appropriate pH is important to ensure compatibility with oral tissues and to prevent enamel damage or irritation. A small amount of the toothpaste (about 1 g) is dispersed in 10 mL of distilled water to prepare a uniform suspension. A strip of pH paper is then dipped into the solution and removed immediately. The color change of the pH paper is compared with the standard color chart provided to determine the pH value [12].

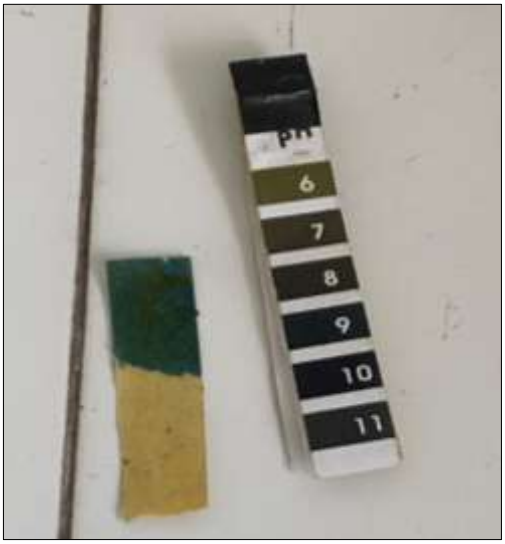


Fig 10: pH paper test

2.4.2 Irritancy test

The irritation test is carried out to evaluate the safety of the toothpaste formulation on oral mucosal tissues. It helps determine whether the product causes any adverse reactions such as redness, inflammation, or discomfort upon application. This test is essential to ensure that the formulation is safe for regular use.

A small amount of the toothpaste is applied on the skin.The area is observed for a specific period for any signs of irritation such as redness, swelling, itching, or burning sensation. In some studies, repeated application may be performed to assess cumulative effects [13].

The subject underwent testing upon providing informed consent.



Fig 11: Irritation test

2.4.3 Organoleptic evaluation

Organoleptic evaluation refers to the assessment of a formulation based on sensory and visual characteristics such as color, odor, taste, and appearance. This test is important because it directly influences consumer acceptability and overall user experienc.

A small quantity of the formulated toothpaste is visually inspected and evaluated by human senses. The parameters assessed include color, odor, taste, and texture/appearance. The observations are recorded based on perception and consistency of the formulation [14].

2.4.4 Stability test

The stability test is performed to assess the ability of the formulated toothpaste to maintain its physical, chemical, and functional properties over time under different environmental conditions. It ensures that the product remains safe, effective, and acceptable throughout its shelf life [16].

2.4.5 Foaming test

The foaming test is performed to determine the ability of a toothpaste to produce foam, which is an important factor for its cleansing action and consumer acceptability. Foam helps in the uniform distribution of the toothpaste in the oral cavity and assists in the removal of food particles and debris. A small quantity of the toothpaste (about 1 g) is taken in a measuring cylinder or test tube containing a fixed volume of distilled water (usually 10-20 mL). The mixture is then shaken vigorously for a specific period (generally 1-2 minutes). After shaking, the solution is allowed to stand, and the volume of foam produced is measured [15].



Fig 12: Foaming test

2.4.6 Viscosity Test

The viscosity test is performed to determine the consistency and flow behavior of the toothpaste formulation. It is an important parameter that affects the ease of extrusion from the tube, spreadability on the toothbrush, and overall consumer acceptability. The viscosity of the prepared toothpaste is measured using a Brookfield viscometer. An appropriate spindle (spindle no. 64) is selected, and the instrument is

operated at a fixed speed of 20 rpm. The toothpaste sample is placed in the viscometer container, ensuring no air bubbles are trapped. The spindle is immersed in the sample, and the reading is recorded once a steady value is obtained. Multiple readings are taken to ensure accuracy [17].

3. Results and Discussion

The formulated polyherbal toothpaste exhibited satisfactory physicochemical and sensory properties, indicating its suitability for oral care use. The pH of the formulation was found to be 7.5, which is close to neutral and considered ideal for maintaining oral health. Such a pH helps prevent enamel erosion and minimizes the risk of irritation to human tissues.

Organoleptic evaluation revealed a pleasant odor, slightly bitter taste, and an olive-green color. The aroma enhances user acceptability, while the slight bitterness is typical of herbal ingredients like neem and clove, which are known for their therapeutic benefits. The natural olive-green color reflects the presence of plant-based components, contributing to the herbal identity of the formulation.

The safety profile of the toothpaste was confirmed through irritation testing, where no redness or irritation was observed. This suggests that the formulation is safe and non-irritant for regular use. Additionally, the foaming test showed adequate foam formation upon agitation, which is essential for effective cleansing. The foaming property is likely due to natural saponins present in herbal ingredients, providing mild surfactant action without the need for synthetic chemicals.

Viscosity measurements using a Brookfield viscometer (spindle #64, 20 rpm) showed values ranging from approximately 5.07 to 5.10 × 10^3 cP. Although this is lower than the typical viscosity range of commercial toothpastes (20-50 × 10^3 cP), it indicates that the paste is smooth and easy to extrude. However, slightly lower viscosity may affect long-term stability.

Overall, the results suggest that the polyherbal toothpaste is safe, effective, and acceptable in terms of sensory and functional properties.

S. No.	Evaluation Test	Standard/Acceptable Value	Observed Result
1	pH paper test	pH should be in the range of 6.5-8.0 for safe oral use	pH = 7.5
2	Irritancy test	No redness, swelling, itching, or irritation should be observed	No redness or irritation observed
3	Organoleptic evaluation - Color	Uniform and acceptable color	Olive-green color
4	Organoleptic evaluation - Odor	Pleasant characteristic odor	Pleasant odor
5	Organoleptic evaluation - Taste	Acceptable pleasant taste	Slightly bitter taste
6	Organoleptic evaluation - Appearance/Texture	Smooth, homogeneous, free from grittiness	Smooth and homogeneous
7	Stability test	No phase separation, color change, odor change, or microbial growth during storage	Stable with no significant change observed
8	Foaming test	Adequate foam formation for effective cleansing	Adequate foam formation observed
9	Viscosity test	Should show good consistency and easy extrusion from tube	5.07-5.10 × 10^3 cP
10	Overall acceptability	Formulation should remain stable, safe, and acceptable	Safe, effective, and acceptable for oral use

4. Conclusion

The plants Neem, Guava, Clove, Cardamom, Turmeric, underwent a series of evaluation tests. The produced compound tested well, yielding favorable findings multiple times. A small group of volunteers has tested the toothpaste and found the to be mouth friendly and does not irritate. Furthermore, the produced toothpaste was standardized through evaluation of various physical and chemical attributes, including pH, appearance, and odor, where the product demonstrates acceptable outcomes.

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References

1. Chugh V, Dhiman S, Mittal V, Singhal A. Formulation and evaluation of herbal toothpaste. Pharmaspire. 2024;16(1):1-2.
2. Khairmode SS, Munde S, Dhole AM, Shinde AA, Shengule AS. Research on formulation and evaluation of herbal toothpaste. International Journal of Scientific Innovation and Engineering. 2025;2(6):355-364.
3. Kumar VS, Navaratnam V. Neem (*Azadirachta indica*): prehistory to contemporary medicinal uses to humankind. Asian Pacific Journal of Tropical Biomedicine. 2013;3(7):505-514.
4. Subapriya R, Nagini S. Medicinal properties of neem leaves: a review. Current Medicinal Chemistry - Anti-Cancer Agents. 2005;5(2):149-156.
5. Petersen PE, Bourgeois D, Ogawa H, Estupinan-Day S, Ndiaye C. The global burden of oral diseases and risks to oral health. Bulletin of the World Health Organization. 2005;83:661-669.
6. Mehta V, Mathur A, Tripathy S, Sa R, Sharma T. Effectiveness of herbal oral care products in reducing dental plaque and gingivitis: an overview of systematic reviews. Canadian Journal of Dental Hygiene. 2024;58(2):120.
7. Bhattacharya T, Saha RP, Majumder R, Ghosh S. Investigating the antimicrobial activity of neem (*Azadirachta indica*) and clove (*Syzygium aromaticum*) extract on biofilm-producing oral microflora. Journal of Experimental Biology and Agricultural Sciences. 2024;12(3):511-521.
8. Suresh S, Arumugham IM, Doraikannan S, Rathinavelu PK, Prabakar J, Balasubramaniam A. Comparing the effectiveness of herbal and conventional dentifrices in reducing dental plaque and gingivitis: a systematic review. Journal of the International Society of Preventive & Community Dentistry. 2021;11(6):601-608.
9. Bodiba DC, Prasad P, Srivastava A, Crampton B, Lall NS. Antibacterial activity of *Azadirachta indica*, *Psidium guajava*, and other medicinal plants and their mechanism of action against *Streptococcus mutans*. Pharmacognosy Magazine. 2018;14(53):76-80.
10. Weatherly LM, Gosse JA. Triclosan exposure, transformation, and human health effects. Journal of Toxicology and Environmental Health, Part B: Critical Reviews. 2017;20(8):447-469.

11. Singh V, Kumar P. Environmental impact of herbal toothpaste: biodegradability and reduced chemical pollution. International Journal of Innovative Research in Technology. 2025;12(7):4579-4585.
12. Shaikh AA. Comparative evaluation of herbal toothpaste formulations: a preliminary study. Indian Journal of Pharmacy and Pharmacology. 2023;10(3):172-176.
13. Skaare A, Kjærheim V, Barkvoll P, Rølla G. Skin reactions and irritation potential of four commercial toothpastes. Acta Odontologica Scandinavica. 1997;55(2):133-136.
14. Ogboji J, Chindo IY, Jauro A, Boryo DEA, Lawal NM. Formulation, physicochemical evaluation and antimicrobial activity of green toothpaste on *Streptococcus mutans*. International Journal of Advanced Chemistry. 2018;6(1):108-113.
15. Oluwasina OO, Akinyemi IA, Oladipo IC. Production of herbal toothpaste: physicochemical, organoleptic and antimicrobial properties. Heliyon. 2023;9(3):e13987.
16. Pokharana M, Vaishnav R, Goyal A, Shrivastava A. Stability testing guidelines of pharmaceutical products. Journal of Drug Delivery and Therapeutics. 2018;8(2).
17. Potanin A, Marron G. Rheological characterization of yield-stress fluids with Brookfield viscometer. Applied Rheology. 2021;31(1):1-9.
18. Tambade S, Hingane LD. Formulation and evaluation of toothpaste from Neem. International Journal for Research in Applied Science and Engineering Technology. 2022;7(6):1250-1258.